

Machine Learning-Based Network Intrusion Detection

Final Capstone Report · UNSW-NB15

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Evidence, evaluation, explainability, bias & fairness, operational implications and
reproducibility

1. Executive Summary

This project investigates whether supervised machine learning can distinguish normal from malicious network traffic using the UNSW-NB15 dataset. The work follows a complete experimental workflow: problem framing, data understanding, EDA, preprocessing, feature engineering, feature selection, PCA, model development, cross-validation, hyperparameter tuning, threshold analysis, independent testing, explainability and critical reflection.

The final selected configuration is a shallow **Decision Tree with max_depth=3 and a classification threshold of 0.50**. On the independent testing dataset, it achieved **100% Recall**, with zero false negatives. At the same time, the test produced a **51.84% False Positive Rate**, corresponding to **19,180 false-positive alerts** among 37,000 legitimate flows.

Central finding: the experiment demonstrates strong malicious-traffic detection on the independent test set, but it also quantifies a substantial alert burden. The results provide evidence for further validation rather than evidence of production readiness.

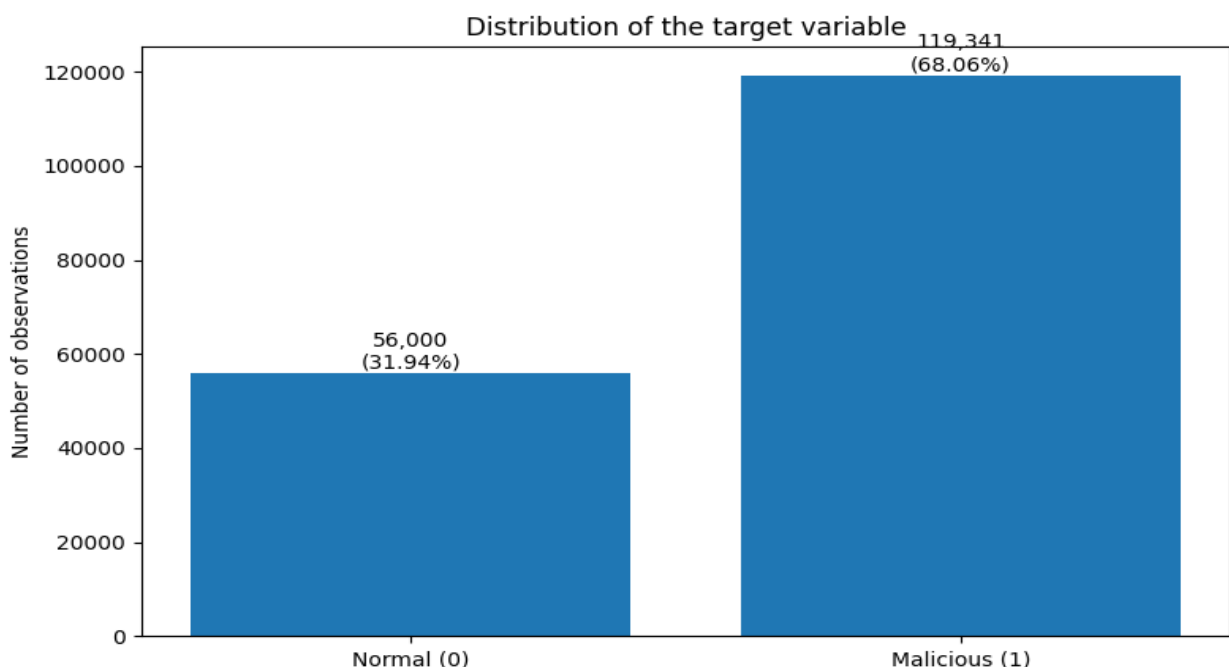
2. Problem & Cybersecurity Context

The business problem is to improve detection of potentially malicious network traffic while maintaining a manageable volume of false alerts. The data-science problem is supervised binary classification of individual network flows: normal (0) versus malicious (1). Recall is the primary detection metric; Precision, F1, PR-AUC, ROC-AUC and FPR provide complementary information.

Scope: this is an academic experimental study of individual network-flow classification. It is not a complete real-time SOC, automated incident-response system or production monitoring platform.

3. Dataset & Data Understanding

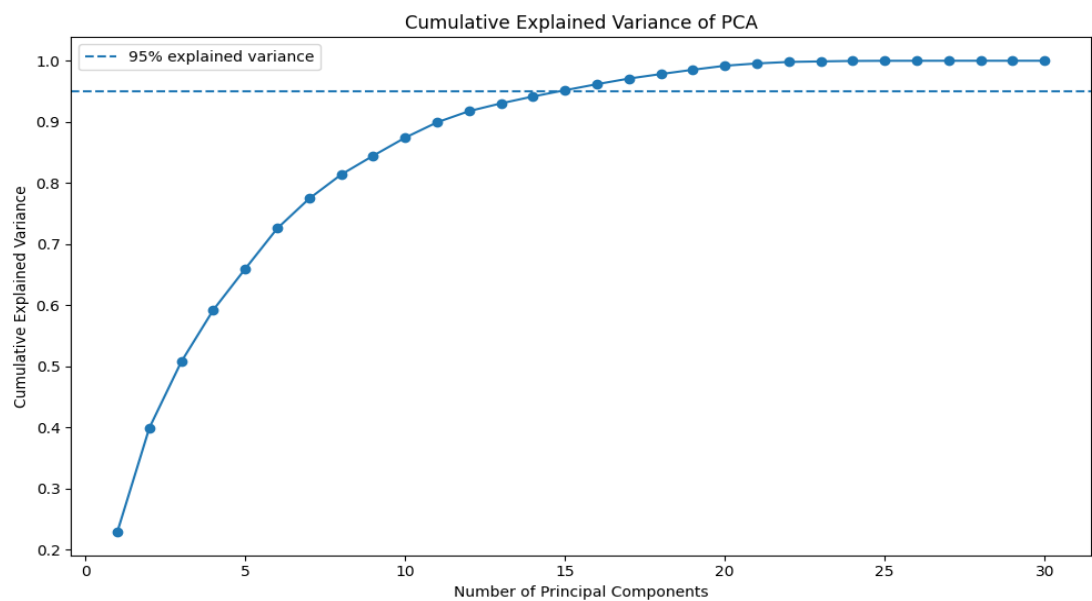
The project uses the UNSW-NB15 dataset. The official training set contains **175,341** observations and the testing set contains **82,332**, for a total of **257,673** observations and 45 columns. The training data contains approximately 68.06% malicious and 31.94% normal observations.



Target distribution.

4. EDA, Preprocessing & Feature Engineering

The EDA examines target distribution, numerical relationships, categorical relationships and correlations. The workflow also performs data-quality checks, categorical encoding, numerical scaling where required, feature engineering, Mutual Information + SelectKBest feature selection and PCA. Three deterministic network-derived features are engineered: total_bytes, total_packets and bytes_ratio. PCA reduces the selected representation to 15 components while retaining approximately 95.18% cumulative explained variance.



PCA cumulative explained variance.

5. Model Development, Evaluation & Selection

Four supervised models are evaluated with Stratified 5-Fold Cross-Validation: Logistic Regression, Decision Tree, Random Forest and XGBoost. Preprocessing is fitted within the modelling pipelines. The independent test set remains separate from model development and threshold selection.

Model	Acc.	Recall	Prec.	F1	PR-AUC	ROC-AUC	FPR
Logistic Regression	93.53%	98.79%	92.25%	95.41%	99.17%	98.42%	17.70%
Decision Tree	94.87%	96.16%	96.30%	96.23%	95.34%	94.30%	7.88%
Random Forest	96.01%	97.88%	96.31%	97.09%	99.69%	99.39%	7.99%
XGBoost	95.77%	97.73%	96.12%	96.92%	99.69%	99.34%	8.40%

Hyperparameter tuning is performed for each model. Threshold analysis uses out-of-fold probabilities from the training data. Maximum OOF Recall was 1.0 and the selected threshold was 0.50.

6. Final Model & Independent Test Evaluation

The final configuration is a Decision Tree with max_depth=3 and threshold 0.50. The model is refit on the complete training dataset and evaluated once on the independent testing dataset.

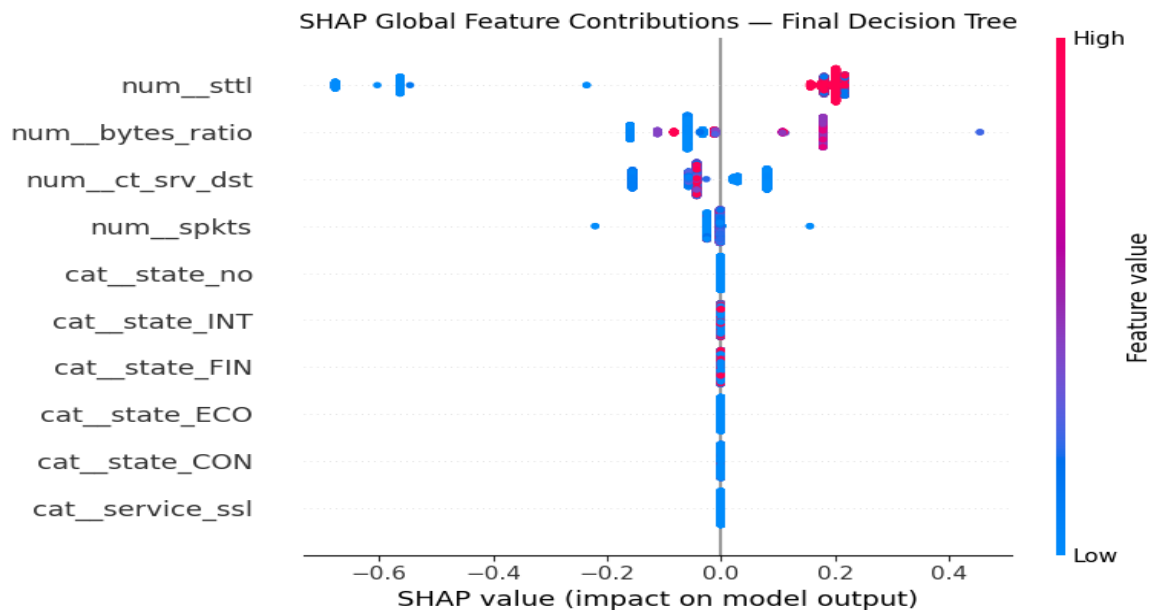
Metric	Independent test
Accuracy	76.70%
Recall	100.00%
Precision	70.27%
F1-score	82.54%
PR-AUC	92.28%
ROC-AUC	93.28%

Metric	Independent test
False Positive Rate	51.84%

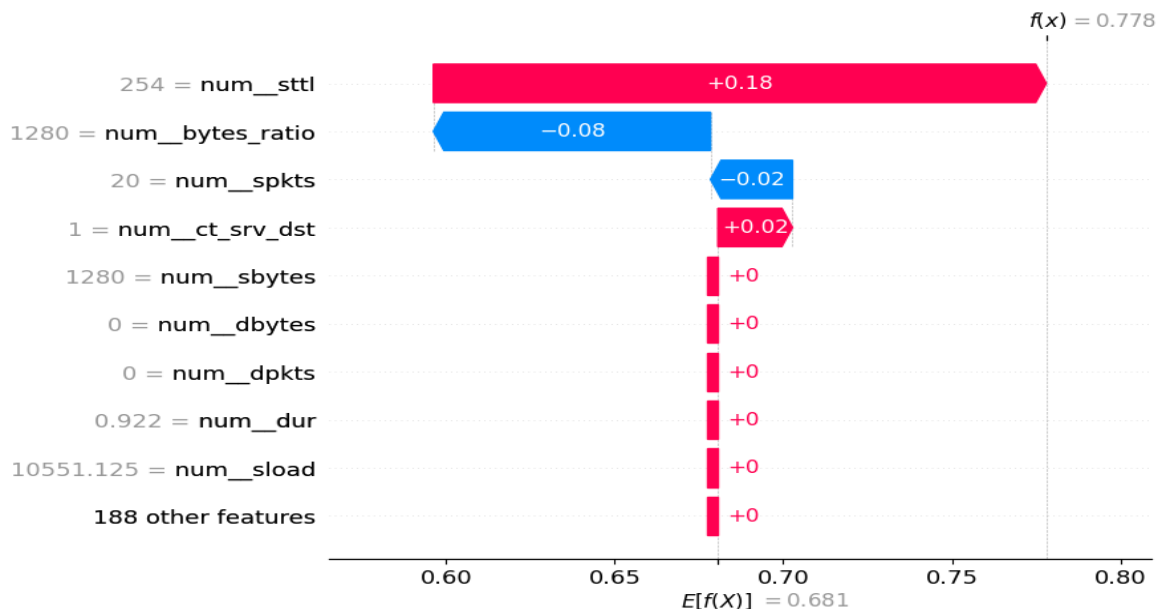
The model detected all malicious observations in the independent test set. However, the 51.84% FPR corresponds to 19,180 legitimate observations incorrectly classified as malicious, or approximately 2,329.59 false alerts per 10,000 flows.

7. Explainability

The final Decision Tree is relatively interpretable. sttl accounts for approximately 88.26% of impurity-based feature importance, followed by ct_srv_dst and bytes_ratio. SHAP is used for global and local inspection of feature contributions. SHAP explains model behaviour; it does not establish causality.



SHAP global explanation.



8. Bias & Fairness Analysis

The UNSW-NB15 dataset does not contain demographic or protected attributes normally used for traditional demographic fairness assessment. The project therefore does not claim to measure demographic fairness. Instead, it examines

whether detection behaviour changes across categories of network traffic: protocol, service and connection state.

The audit is performed on the independent test set using the final model and threshold. Groups with at least 500 observations are retained for the main audit.

Group	Support	Recall	Precision	F1	FPR
TCP	43,095	100.0%	49.35%	66.08%	56.20%
UDP	29,418	100.0%	85.90%	92.41%	43.23%
HTTP	8,287	100.0%	64.41%	78.35%	58.86%
FIN	39,339	100.0%	55.69%	71.54%	49.92%
INT	34,163	100.0%	89.41%	94.41%	78.37%
CON	6,982	100.0%	15.33%	26.59%	29.05%
REQ	1,842	100.0%	7.48%	13.92%	97.83%

The audit identifies variation in error behaviour across traffic groups. These differences do not demonstrate demographic bias; they indicate that false-positive and classification behaviour varies by traffic type and may therefore have operational consequences.

9. Operational Implications, Reliability & Limitations

Detection effectiveness: 100% Recall and 0% undetected attack rate on the independent test. **Alert burden:** 51.84% FPR, 19,180 false-positive alerts and approximately 2,330 per 10,000 flows.

The independent test also shows a decrease in several metrics compared with cross-validation during model development. The dataset represents a specific experimental environment and does not establish performance on other datasets, live traffic or a production SOC. Explainability and subgroup analysis improve inspection but do not establish operational reliability.

Potential mitigation directions are more representative data, continued model/configuration evaluation, threshold adjustment and human-in-the-loop review. These measures require validation in the target environment.

10. Reproducibility & GitHub

Public repository: <https://github.com/valyry-bot/unsw-nb15-intrusion-detection>

Reproducible code: the final Jupyter notebook documents the complete workflow, including data preparation, EDA, feature engineering, model development, evaluation, explainability and critical analysis.

Notebook: `notebooks/Valerie_Andoulsi_Capstone_Machine_learning_based_network_intrusion_detection.ipynb`

The repository contains the final model artefact and configuration, project documentation and the EDA/feature-engineering report. The raw UNSW-NB15 dataset is not redistributed.

11. Conclusion

The project demonstrates that machine learning can distinguish normal and malicious network traffic within the UNSW-NB15 experimental setting. The final Decision Tree achieved 100% Recall on the independent test set, but this was accompanied by a 51.84% False Positive Rate and a substantial number of false-positive alerts.

The central result is therefore the measured detection/alert trade-off. Before any operational use, validation on representative target-environment traffic, assessment of analyst workload and continued human oversight would be required. The project establishes an evidence base for validation rather than a production-ready cybersecurity solution.